

MUDIT ATREY

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EDUCATION

Kanpur Institute of Technology, AKTU

Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization)

Kanpur, India

2023 – June 2027

- **Academic Standing:** CGPA 7.3/10.0; Core Focus: Machine Learning, Numerical Methods, Scientific Computing.
- **Availability:** Available for full 13-month industrial placement starting July 2027 as an intentional pre-postgraduate placement year.

RESEARCH & AERODYNAMIC SIMULATION

Physics-Regularized Geometric Manifolds for Surrogate-Assisted MDO of Tailless BWB UAVs

Aug 2026

- **Publication Status:** First-author preprint (DOI: [10.5281/zenodo.22893593](https://doi.org/10.5281/zenodo.22893593)); under external peer review at *Aerospace Science and Technology* (Elsevier, IF 6.4, Q1), Manuscript ID: AESCTE-D-26-06017.
- Formulated C¹-continuous Hermite geometric manifold and active-learning 5-MLP surrogate ensemble with mass-centroid balancing, cutting solver calls by 80%+ across 50,000+ VLM evaluations compared to Latin Hypercube Sampling (AVL, XFOIL, AeroSandbox, OpenVSP).
- **High-Fidelity 3D RANS CFD:** Executed 3D incompressible RANS in SimScale cross-validated against SU2 (Spalart-Allmaras); identified 124% parasite drag rise (CD0 = 0.0242, peak L/D = 13.1) over low-fidelity polar drag build-up from 0.84 m² turbulent wetted area at Re = 800k.
- **OpenCASCADE B-Rep Pipeline:** Built custom B-Rep lofting pipeline on OpenCASCADE to eliminate micro-gaps and zero-thickness trailing edges from polygon-to-NURBS mesh translations, generating certified watertight solids for CFD meshing.

EXPERIENCE

Pradite (pradite.com)

Founder & Technical Lead

Kanpur, India

Aug 2026 – Present

- Founded zero-knowledge BYOK AI coding platform; led 5-person engineering team through full-stack architecture, build, and Early Access launch.
- Engineered zero-trust security core: client-side XChaCha20-Poly1305 + Argon2id encryption (libsodium), sub-5ms encrypted IndexedDB index search, and direct client-to-model streaming with zero server-side storage.
- Built multi-provider inference engine with dynamic complexity routing across 17 APIs/runtimes (vLLM, Ollama); filed Class 42 trademark (IP India) and executed pre-incorporation co-founders agreement.

OPEN SOURCE ENGINEERING

scikit-learn Ecosystem (proabl-ai / skrub-data)

Open Source Contributor (ESoC)

Open Source

2025 – Present

- **skrub:** Refactored CheckInput pipeline logic to bypass eager evaluation of Polars lazyframes, cutting memory overhead during dataset ingestion.
- **skore:** Implemented multi-metric aggregate parameters for ImpurityDecreaseDisplay and CoefficientsDisplay, enabling flexible model explainability for tree and linear ensembles.
- Fixed figure-level legend persistence during file export in PredictionErrorDisplay; 5 merged PRs reviewed and approved by core maintainers (@glemaitre, @rcap107, @GaetandeCast).

PROJECTS

LLM-Assisted Parametric CAD Framework (onshape-mcp) | Python, Onshape REST API, MCP

2026

- Architected an agentic CAD system driving programmatic 3D geometry creation and parametric editing in Onshape via REST API from natural language specifications.
- Maintains bidirectional state tracking of geometric features, enforces topological constraints, and plans multi-step CAD feature trees derived from Onshape API schemas.

6-DOF Aerodynamic Guidance & Trajectory Simulation (SIH26) | Python, 6-DOF, EKF, Onshape

2026

- Modeled 6-DOF trajectory simulation and Augmented Proportional Navigation (APN) guidance law for 155mm munition, targeting $\leq 30\text{m}$ CEP over 20km+ ballistic arc (Ministry of Defence / SIH26098).
- Implemented 16-state Extended Kalman Filter (EKF) fusing 500Hz IMU with 10Hz GNSS; designed cruciform canard array decoupling passive roll despin from pure lateral steering force.

TECHNICAL SKILLS & INTERESTS

Languages:

Python, C++, Rust, TypeScript, JavaScript, SQL, HTML/CSS

Aero, CFD & CAD:

SimScale (RANS), SU2, OpenCASCADE, Onshape (API & CAD), AVL 3.27, XFOIL, AeroSandbox, OpenVSP

Data Science & ML:

scikit-learn, Optuna, Pandas, NumPy, SciPy, MLX, PyTorch, skrub, skore

Systems & Tools:

Git, GitLab CI/CD, Docker, Linux, WebAssembly, Supabase, Cloudflare Pages, PlatformIO, ESP8266

Interests:

Formula 1, Chess (~1600 ELO), Embedded Systems, Hardware Builds